“What Is the Effect of UN Involvement on Post-War Peace?”
We Have Different Answers - Which Do We Trust?

Extending Cross-Validation Logic to Estimate Out-of-Convex Hull Prediction

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Whenever we look back at history, in addition to getting the fuzzy feeling of nostalgia, usually we want to understand. Why did things happen? Why did an initial event lead to subsequent events? What was the underlying cause? What were the underlying rules in play?

In addition to understanding, sometimes we also want to juxtapose. We want to rearrange. How different would the present be if history was different in some way? How different would today be if event X had happened yesterday instead of event Y? In asking these questions, we explore counterfactuals - possibilities separate from reality.

We use various tools for this exploration. In many cases we use our brains. Sometimes we use computers. With computers we can explore history at a scale that used to be practically impossible. We can analyze huge amounts of data and draw insightful inferences from them.

Back to juxtaposition - a recent paper by King[1] showed that there are limits on how much juxtaposition we can reliably carry out. Sometimes these possibilities - these counterfactual cases are so different from actual instances in reality that we cannot trust our computer models to reliably predict them. In such situations, whatever insights our models give us are more reflective of the type of models we use, than the actual laws underlying reality.

King's analysis of a study by Doyle[2] showed that the causal claims made about the effect of UN intervention on post-war peace made use of prediction for extreme counterfactuals — which as we have just seen is unreliable. How then do we reliably estimate such undoubtedly important effects?

Below we view a diagram from King's paper[1]. The diagram below clearly describes this phenomenon of model dependence. We see that for an X value of 5 (possibly representing some distant counterfactual case), whatever prediction we have seems to be entirely dependent on our choice of model. At that distance, our prediction has practically nothing to do with the existing data.

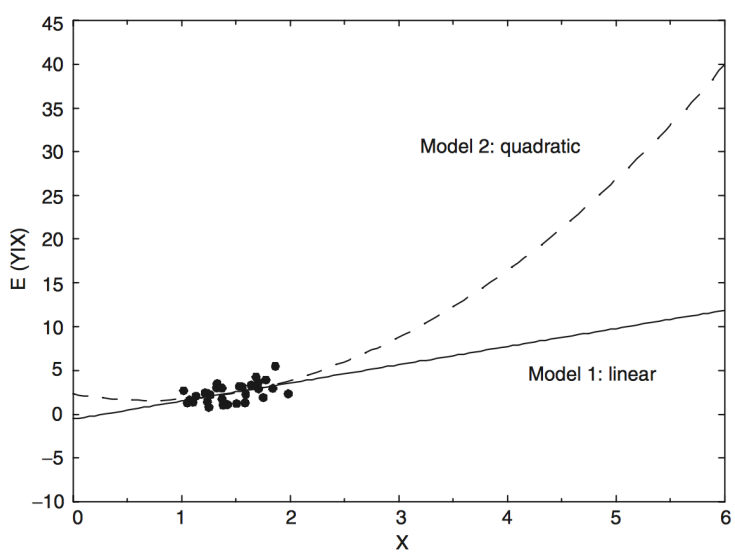


FIG. 1. Linear and Quadratic Models With Equal Fit to Simulated Data But Massively Different Out-of-Sample Implications

This phenomenon is what makes prediction for distant counterfactuals unreliable, and thus places limits on its utility.
In this essay we explore a possibility. We explore the possibility that although inference is generally unreliable for distant counterfactual cases, we might be able to determine which models are closer to the truth than others. We might be able to determine which models to trust more than others.

In exploring this possibility, we attempt to broaden the discussed limits on reliable causal inference.

Approach:

For our approach, we draw insight from the technique of cross-validation. In cross-validation we aim to estimate how well a model trained on some dataset X_1, will perform on some other dataset X_2, where X_1 and X_2 are drawn from the same population.

The logic underlying this aims to test for model dependence. Essentially we are testing how well the model generalizes to X_2. We are testing how well the model trained on X_1, learns the rules of the underlying population distribution. If this model learns the underlying rules well, then it should have a satisfactory performance on X_2. However if it learns something else - something different from these underlying rules, then we expect it's performance on X_2 to be relatively poor.

This idea connects to the perspective in the introduction about history, reality and counterfactuals. We can view cases in reality as our X_1. We can view the extreme counterfactual cases as our X_2. In contrast to cross-validation however, we are certain that the points in X_2 are external and "far away" from the ones in X_1. I.e., in this case there is a notion of distance, unlike in cross validation which only assumes two different samples drawn from the same population.

Here we attempt to extend this idea behind cross validation and define it such that it takes into consideration the notion of distance.

The Gower Metric G^2

King used two different metrics to describe how "distant and isolated" a counterfactual was from the rest of the data. The first metric was the convex hull. In this metric, a point x is "distant" from a set of points X, if x exists outside the convex hull of X.

The second (and finer) metric was the Gower metric G^2. G^2 is a number ranging from 0 upwards, that describes how far a point x_1 is from another point x_2, relative to the spread of a set of values X.

$$G^2 = \frac{|x_1 - x_2|}{max(X) - min(X)}$$

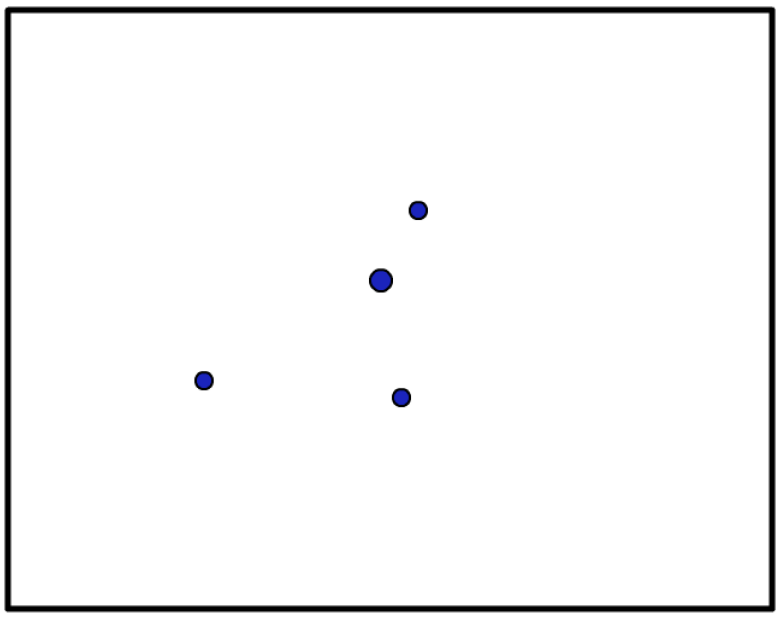
(The one-dimensional case is shown here for simplicity)

Thus given our historical data X, and some point x_2 representing some distant counterfactual, the G^2 for some x_1 (drawn from X) and x_2, describes how “far away” x_2 is from x_1, relative to the spread of values in X.

In our approach, we use the Gower metric to define our notion of distance.

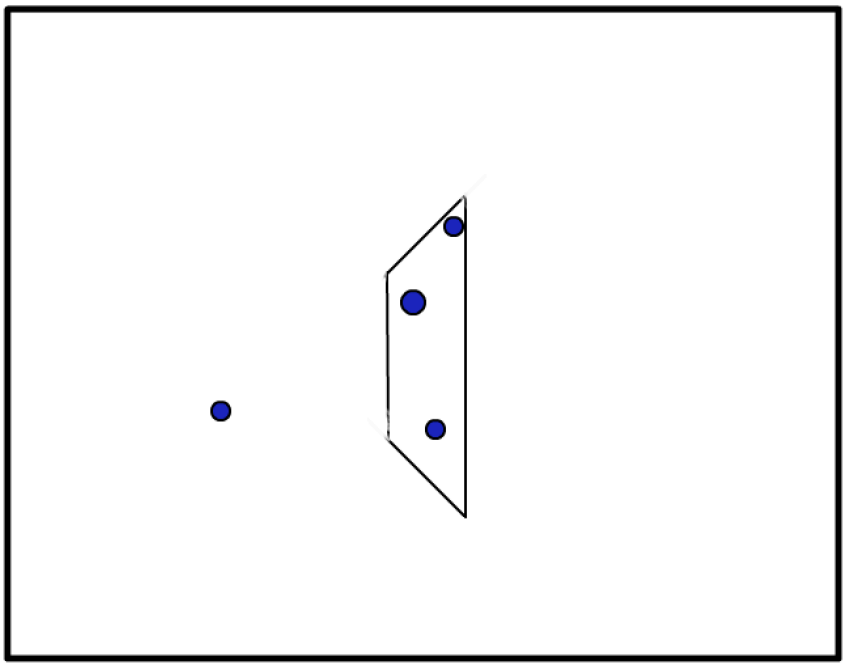
<u>Convex Hull Iteration:</u>

Say we have a given set X, containing data points in two dimensions. A convex hull iteration through X begins from the central element in X, and then proceeds outwards element-wise, until it reaches the outermost element of X. We illustrate this below:



The diagram above represents X. We can see the 4 different points contained in X. The first step of a convex hull iteration starts at the central element of X. The second step expands to the second central element, etc.

Thus the third step of the iteration will look like this:



We see that essentially the process involves iterating outwards through convex hulls spreading throughout X.

<u>Application of Convex Hull Iteration:</u>

Say we train a prediction model on the 3 innermost points of X - the 3 elements within the convex hull illustrated above. We could test this model on the fourth point - the point just outside the hull in the diagram. Since we have complete information about this data point, we can instantly verify if our model's prediction was correct or not.

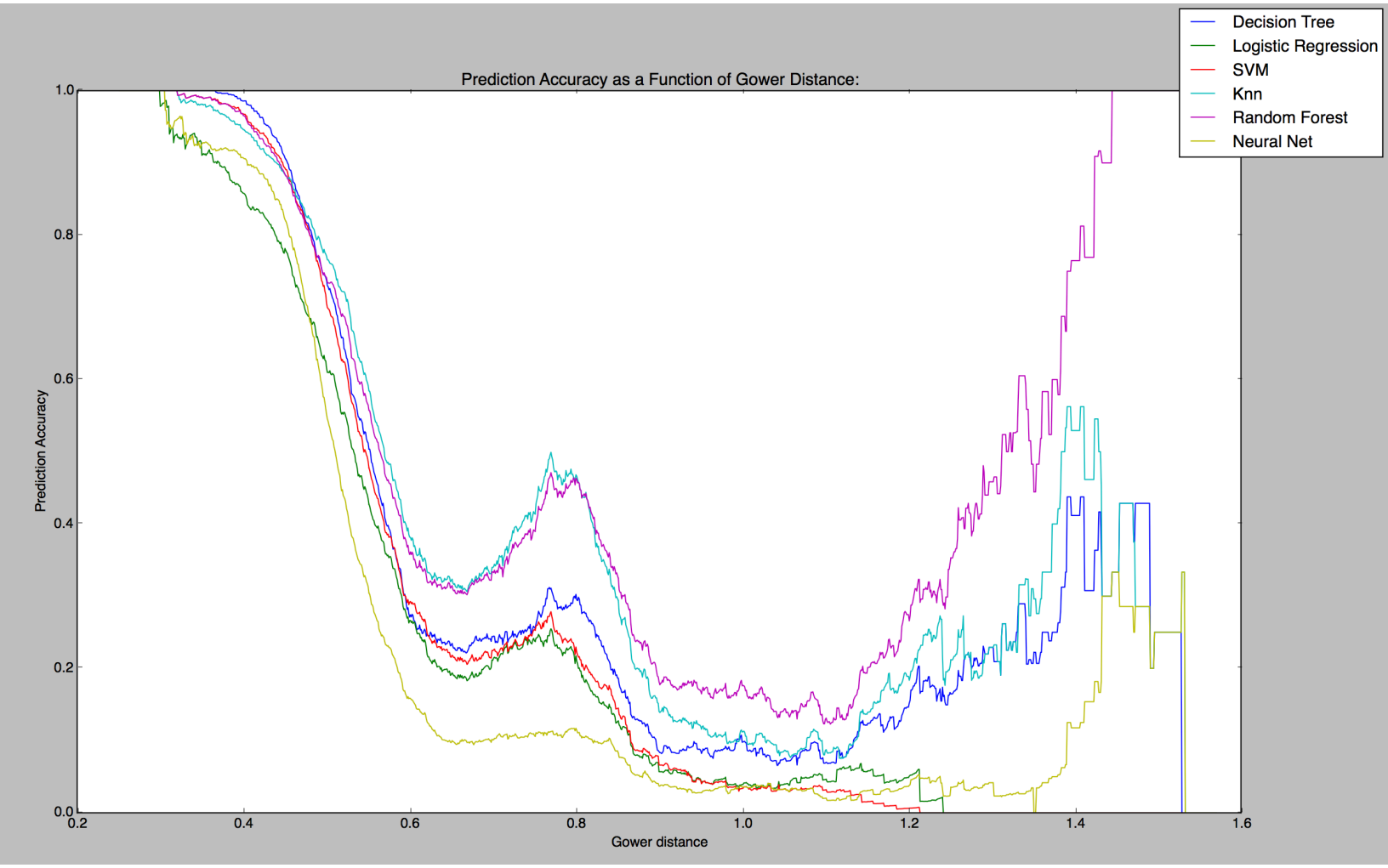
Recalling the notion of distance — the Gower metric. We can calculate the Gower distance G^2 between this fourth point, and the central element of X. This quantifies our notion of distance as G^2 tells us how "far away" from the training set our fourth point is.

At this point we have two quantities. The prediction accuracy of our model for this fourth point, and at what distance we observed this accuracy.

We extend this logic: We train a given model on the n-innermost elements (where n ranges from some baseline to the total number of elements) of our dataset, and for the remaining elements, we calculate the prediction accuracy of our model, and at what distance this prediction accuracy was observed.

Similar to how cross-validation offers performance on subsets of the training data as an estimate for out of sample performance, this approach offers performance on external points (points outside the convex hull of some subset of the training data), as an estimate of performance on points external to the entire training data.

Analysis:
After describing the above technique in code[3], running with different models on the popular Iris dataset gave the results shown below:



In the diagram above, we see that all the model curves tend to follow a characteristic trend: For data points very close to the training data they perform optimally. Then as these points get farther away, the performance drops quickly. Around G^2 of about 0.8 however, there is a significant performance boost. The downward trend continues again and for some of the models it drops to zero. For some others however, there is another boost in performance, even going up to 1 in the case of the random forest.

It will be interesting to explore these curves and understand why they have the observed shape.

The key observation relating to our aim here however, is that although the different curves have largely the same shape, some models can be seen to perform consistently better than others. The random forest model for example, consistently outperforms the support vector machine.

We have discussed how prediction is unreliable for points distant from the training data. However this notion that some models will consistently outperform some others at this task - a notion separate from the usual training set performance (all the models in the figure perform optimally for short enough Gower distances), this notion will be key in determining what model, of a given number of models, will be the most reliable in predicting for extreme data points.

It will be a useful and interesting endeavor to explore this approach more, apply it to more models and more datasets, and more deeply understand its implications.

So although the insights in Gary's paper shed light on the fact that counterfactual prediction and consequently causal inference is unreliable for far-off counterfactuals, the approach presented in this essay can help determine what model is the most likely of a given set of models, to predict correctly for extreme counterfactuals. In this way we hope to broaden the present limits on extreme counterfactual prediction and inch closer to more reliable, insightful and interesting causal inference.

References:

1. DOYLE, MICHAEL W., AND NICHOLAS SAMBANIS. (2000) International Peacebuilding. American Political Science Review.

2. KING, GARY and ZENG, LANGCHE. (2007) When Can History Be Our Guide? The Pitfalls of Counterfactual Inference.

3. Code for Convex Hull Iteration: https://github.com/mayowaosibodu/Convex-Hull-Iteration